

BCSE-011: Fundamental of Artificial Intelligence & Machine Learning

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UNIT I (25%)

AI Foundations, Agent Architectures, PEAS, Turing Test, Ethics & Modern Applications.

UNIT II (25%)

State Space Search (BFS, DFS, A*), Knowledge Representation, FOPC & Resolution.

UNIT III (25%)

Expert Systems, Inference Engines, Knowledge Engineering & Case Studies (MYCIN/DENDRAL).

UNIT IV (25%)

Machine Learning, Supervised vs Unsupervised, Regression, Decision Trees, Metrics.

UNIT I: FOUNDATIONS OF ARTIFICIAL INTELLIGENCE & INTELLIGENT AGENTS

Evolution, Definitions, Branches, AI vs ML vs DL, Agent Paradigms, Turing Test & Ethics

1. Definition, Evolution & Branches of Artificial Intelligence

Definition: Artificial Intelligence is the subfield of computer science dedicated to creating computational systems capable of performing tasks that typically require human cognition—such as visual perception, natural language understanding, automated reasoning, knowledge representation, and adaptive decision-making.

Evolution Timeline:

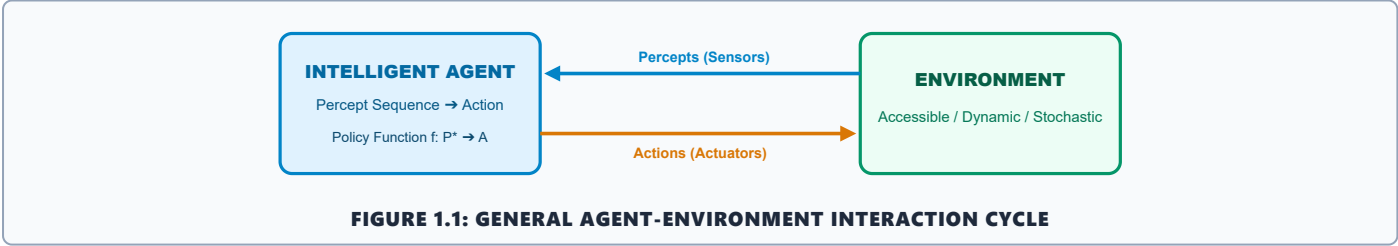
- **1950 (Foundational Dawn):** Alan Turing publishes *Computing Machinery and Intelligence*, formulating the operational Turing Test.
- **1956 (Dartmouth Workshop):** John McCarthy coins the term "Artificial Intelligence" alongside Marvin Minsky, Claude Shannon, and Nathaniel Rochester.
- **1965–1975 (Symbolic Reasoning & Micro-worlds):** Birth of ELIZA (first chatbot) and SHRDLU (blocks world).
- **1980s (The Expert System Boom):** Rule-based production systems (XCON, MYCIN) revolutionize industrial diagnosis and commercial deployment.
- **1997 (Deep Blue):** IBM Deep Blue defeats world chess champion Garry Kasparov via massively parallel alpha-beta pruning.
- **2012–Present (Deep Learning & GenAI Renaissance):** AlexNet revolutionizes computer vision using GPUs; Transformers and Large Language Models (LLMs) achieve superhuman benchmarks in multimodal synthesis.

2. Comprehensive 12-Point Comparative Matrix: AI vs. Machine Learning vs. Deep Learning

Dimension	Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)
1. Conceptual Scope	The superset of all machines mimicking human intelligence and rational thought.	A specialized branch of AI learning patterns statistically from data.	A specialized subset of ML based on deep multi-layered neural networks.
2. Historical Emergence	Formally established at Dartmouth Conference in 1956.	Surged in 1980s–1990s as statistical and probabilistic learning.	Accelerated in 2012 following ImageNet and GPU computational scaling.
3. Core Mechanism	Symbolic logic, production rules, heuristic search, statistical learning.	Mathematical statistical models and optimization algorithms (cost minimization).	Backpropagation through artificial neural networks with multiple hidden layers.
4. Feature Engineering	Manually encoded by human engineers via knowledge bases and ontologies.	Requires expert manual feature extraction (e.g. SIFT, HOG, PCA).	Automated end-to-end representation learning directly from raw data tensors.
5. Data Appetite	Can operate with zero data using purely hand-crafted symbolic rules.	Excels on small-to-medium tabular datasets (thousands of samples).	Data-hungry; requires massive scale (millions of samples) to avoid overfitting.
6. Computational Load	Runs on standard CPU hardware architectures.	Moderate computational load; trains quickly on standard multi-core CPUs.	Massive computational requirements; necessitates dedicated GPUs/TPUs.
7. Interpretability	High transparency in symbolic/rule-based systems; white-box reasoning.	Moderate interpretability (coefficients in regression, paths in decision trees).	Black-box behavior; complex high-dimensional parameter spaces (XAI required).
8. Performance Curve	Static performance bound strictly by pre-programmed knowledge capacity.	Plateaus after reaching a moderate data volume threshold.	Continuous performance scaling as data volume and model parameters increase.
9. Hardware Dependency	General purpose x86/ARM microprocessors.	Standard CPU clusters with moderate RAM memory footprints.	Parallel tensor cores (NVIDIA CUDA, Google TPU, Apple Neural Engine).
10. Problem Modality	Structured theorem proving, state-space puzzle solving, expert rules.	Structured tabular data, customer churn, fraud scoring, house pricing.	Unstructured sensory data: high-res vision, continuous audio, raw text.
11. Failure Modes	Brittle out-of-distribution failure when encountering unmapped states.	Underfitting or severe overfitting on unrepresentative sample bias.	Hallucination, catastrophic forgetting, adversarial pixel attacks.
12. Canonical Examples	Deep Blue, Rule-based medical diagnostic systems, A* route planning.	Linear Regression, Random Forests, Support Vector Machines (SVM), k-NN.	Convolutional Neural Networks (CNN), Transformers (BERT, GPT-4), ResNet.

3. Intelligent Agents, PEAS Framework & 5 Core Agent Classes

An **Agent** is anything that can perceive its environment through **Sensors** and act upon that environment through **Actuators**. The rational agent always acts to maximize its expected performance measure given its percept history.



PEAS Breakdown (Autonomous Taxi Example):

- Performance Measure:** Safety, passenger comfort, travel speed, legal compliance, fuel economy.
- Environment:** City streets, pedestrians, highways, erratic drivers, diverse weather conditions.
- Actuators:** Steering wheel, throttle accelerator, brakes, horn, turn signals.
- Sensors:** Cameras, LiDAR, radar, ultrasonic sonar, GPS, speedometer, accelerometer.

UNIT II: PROBLEM SOLVING & KNOWLEDGE REPRESENTATION

State Space Formulation, Uninformed vs Informed Search, A* Optimality, FOPC & Resolution Refutation

1. State Space Search & Heuristic Algorithms

A computational problem is formulated in AI as a 5-tuple: $\{S_0, \text{Actions}(s), \text{Result}(s, a), \text{Goal-Test}(s), \text{Path-Cost } c(s, a, s')\}$.

Algorithm	Queue Strategy	Time Complexity	Space Complexity	Completeness	Optimality
Breadth-First Search (BFS)	FIFO (First-In, First-Out)	$O(b^d)$	$O(b^d)$ [High Memory!]	Yes (if b is finite)	Yes (if step costs equal)
Depth-First Search (DFS)	LIFO (Stack)	$O(b^m)$	$O(b \cdot m)$ [Linear Memory]	No (fails in infinite paths)	No (can find long path)
Uniform Cost Search (UCS)	Priority Queue by $g(n)$	$O(b^{1 + \lceil \frac{C^*}{\epsilon} \rceil})$	$O(b^{1 + \lceil \frac{C^*}{\epsilon} \rceil})$	Yes	Yes (arbitrary costs $\geq \epsilon$)
A* Search	Priority Queue by $f(n) = g(n) + h(n)$	$O(b^d)$	$O(b^d)$	Yes	Yes (if h is admissible)

A* Heuristic Admissibility & Consistency:

- Admissibility:** A heuristic $h(n)$ is admissible if it never overestimates the true cost to reach the goal: $0 \leq h(n) \leq h^*(n)$.
- Consistency (Monotonicity):** For every node n and successor n' generated by action a : $h(n) \leq c(n, a, n') + h(n')$ (satisfies Triangle Inequality). Consistency guarantees A* finds optimal paths without reopening closed nodes!

2. First-Order Predicate Calculus (FOPC) & Resolution Refutation Pipeline

While Propositional Logic cannot express relationships between objects and variables, FOPC introduces Objects, Predicates, Functions, and Quantifiers ($\forall, \exists$).

The Complete 8-Step CNF Conversion Pipeline:

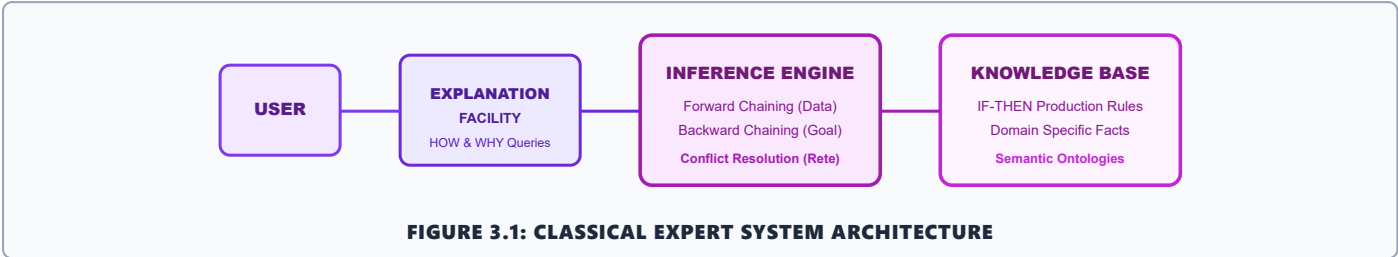
- Eliminate Implications ($\Rightarrow$ and $\Leftrightarrow$):** Replace $A \Rightarrow B$ with $\neg A \vee B$; replace $A \Leftrightarrow B$ with $(\neg A \vee B) \wedge (\neg B \vee A)$.
- Move Negation Inward (NNF):** Apply De Morgan's laws: $\neg(A \wedge B) \equiv \neg A \vee \neg B$ and quantifier duality: $\neg \forall x P(x) \equiv \exists x \neg P(x)$; $\neg \exists x P(x) \equiv \forall x \neg P(x)$.
- Standardize Variables:** Ensure each quantifier binds a unique variable symbol: $(\forall x P(x)) \vee (\forall x Q(x)) \rightarrow (\forall x P(x)) \vee (\forall y Q(y))$.
- Skolemization (Eliminate $\exists$):** Replace existential variables with constants or Skolem functions depending on universally quantified variables: $\forall x \exists y \text{Mother}(y, x) \rightarrow \forall x \text{Mother}(f(x), x)$.
- Drop Universal Quantifiers ($\forall$):** All variables are now implicitly universally quantified.
- Distribute $\vee$ over $\wedge$:** Transform into Conjunctive Normal Form: $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$.
- Clause Segmentation:** Treat each conjunct as a distinct disjunctive clause.
- Standardize Variables Across Clauses:** Rename variables so no two clauses share identical variable names.

UNIT III: EXPERT SYSTEMS & KNOWLEDGE ENGINEERING

Architecture, Inference Strategies, Knowledge Acquisition, Certainty Factors & Classic Case Studies

1. Architecture and Operational Mechanism of Expert Systems

An Expert System is an AI system that emulates the decision-making ability of a human expert in a specialized domain. Its hallmark architectural feature is the strict separation of domain knowledge (Knowledge Base) from general reasoning logic (Inference Engine).



Component	Function & Role	Practical Engineering Challenge
Knowledge Base (KB)	Stores domain-specific declarative assertions and heuristic IF-THEN production rules.	The "Knowledge Acquisition Bottleneck"—extracting tacit expertise from human experts.
Inference Engine	Applies rules of deduction to working memory to derive fresh facts and solve problems.	Computational explosion as rule base scales; requires Rete algorithm optimizations.
Working Memory	Dynamic global database holding case-specific observations, symptoms, and intermediate deductions.	Maintaining cache coherence and state cleanup across multi-step diagnosis.
Explanation Facility	Traces execution history to answer "WHY" a question is prompted and "HOW" conclusions were inferred.	Generating human-readable plain language narratives rather than raw logic traces.

2. Forward vs. Backward Chaining & Benchmark Historical Case Studies

Comparison Dimension	Forward Chaining (Data-Driven)	Backward Chaining (Goal-Driven)
Inference Direction	Starts with known atomic facts → applies Modus Ponens → infers new conclusions.	Starts with a proposed Hypothesis/Goal → searches backward for supporting facts.
Optimal Use Case	Monitoring, process control, automated synthesis, real-time alerting.	Medical diagnosis, debugging, legal deliberation, forensic investigation.
Rule Activation Strategy	Fires every rule whose antecedent (IF) matches current working memory.	Selects rules whose consequent (THEN) matches current target sub-goal.

Historical Landmark Case Studies:

- MYCIN (Stanford, 1970s):** Developed by Edward Shortliffe to diagnose blood-borne infectious diseases (bacteremia and meningitis). Incorporated **Certainty Factors (CF $\in [-1.0, +1.0]$)** to handle uncertain medical evidence.
- DENDRAL (Stanford, 1965):** Led by Edward Feigenbaum; inferred organic chemical molecular structures from mass spectrometer data using heuristic pruning.

UNIT IV: INTRODUCTION TO MACHINE LEARNING & EVALUATION METRICS

Taxonomy, Supervised (Regression, Trees, Bayes, k-NN), Unsupervised (k-Means, Apriori) & Performance Scoring

1. Taxonomy of Machine Learning Paradigms

Paradigm	Input Data Format	Optimization Objective	Representative Algorithms
Supervised Learning	Labeled pairs $\{(x_i, y_i)\}$ with explicit ground truth.	Minimize Empirical Risk (Loss Function): $L = \sum y_i - f(x_i) ^2$.	Linear/Logistic Regression, Decision Trees (ID3/C4.5), Naive Bayes, k-NN, SVM.
Unsupervised Learning	Unlabeled dataset $\{x_i\}$ without external supervision.	Discover intrinsic manifold geometry, clustering density, or association co-occurrence.	k-Means Clustering, Agglomerative Hierarchical, PCA, Apriori Association.
Reinforcement Learning	Sequential states s_t , actions a_t , scalar rewards r_t .	Maximize cumulative expected discounted reward: $R = \sum \gamma^t r_t$.	Q-Learning, SARSA, Policy Gradients, Actor-Critic (PPO, DDPG).

2. Mathematical Formulation of Core Algorithms

1. Decision Tree (ID3) Splitting Criterion:

- Entropy of Sample Set S : $\text{Entropy}(S) = - \sum_{i=1}^c p_i \cdot \log_2(p_i)$
 - Information Gain of Attribute A : $\text{Gain}(S, A) = \text{Entropy}(S) - \sum_{v \in \text{Values}(A)} (|S_v| / |S|) \cdot \text{Entropy}(S_v)$
- Selection Rule: Select the attribute that maximizes Information Gain at each tree node.

2. Naive Bayes Classification:

$$P(C_k | X) = \frac{P(C_k) \cdot \prod_{j=1}^d P(x_j | C_k)}{P}(X)$$

Conditional Independence Assumption: Assumes all features x_j are mutually independent given class label C_k .

3. k-Means Clustering Algorithm:

- Choose k cluster centroids randomly from dataset: $\mu_1, \mu_2, \dots, \mu_k$.
- Assignment Step:** Assign each data point x_i to the nearest centroid using Euclidean distance: $c_i = \operatorname{argmin}_j ||x_i - \mu_j||^2$.
- Update Step:** Recompute the centroid of each cluster as the arithmetic mean of all assigned points: $\mu_j = (1 / |C_j|) \sum_{x_i \in C_j} x_i$.
- Repeat Steps 2 and 3 until centroids stabilize (convergence condition: $\Delta\mu < \epsilon$).

3. Confusion Matrix, Error Types & Comprehensive Metric Formulation

Metric Name	Mathematical Formula	Strategic Importance & Exam Insight
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	General sanity metric; severely biased in class-imbalanced contexts (e.g. 99% non-fraud).
Precision	$TP / (TP + FP)$	Measures reliability of positive predictions. Critical when False Positive cost is high (Email spam).
Recall (Sensitivity)	$TP / (TP + FN)$	Measures ability to find all positive instances. Critical when False Negative is lethal (Cancer detection).
Specificity	$TN / (TN + FP)$	True Negative Rate; measures ability to identify actual negatives.
F1-Score	$2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$	Harmonic mean balancing Precision and Recall; robust under extreme class imbalance.

MMDU END-SEMESTER EXAMINATION PAPER (SOLVED WITH DUAL DIAGRAMS)

Batch 2025–2029 • BCSE-011: Fundamental of AI & ML • Total Marks: 60 • 100% Dual-Diagram Compliance

SECTION A: 12 COMPULSORY CONCEPTUAL QUESTIONS (1 MARK EACH = 12 MARKS)

Question 1 (Unit 1): The Chinese Room Thought Experiment vs. Turing Test

1 Mark

Explain John Searle's Chinese Room argument in refuting the Turing Test as a measure of true machine consciousness.

Question Thought Experiment Setup: Input (Chinese Glyphs) → Room [Human + English Rulebook] → Output (Syntactically Correct Glyphs).

STEP-BY-STEP SOLUTION:

Solution Syntax vs Semantics Bridge:

Symbol Manipulation (Syntax) ≠ Understanding / Intentionality (Semantics).

The person inside passes the Turing test externally without understanding a single word of Chinese.

Searle demonstrated that passing the Turing Test requires only syntactic symbol manipulation, which computers execute effortlessly. True understanding requires semantic intentionality, proving that behavioral mimicry does not equate to cognitive consciousness.

Question 2 (Unit 1): Rationality vs. Omniscience in AI Agents

1 Mark

Distinguish between a Rational Agent and an Omniscient Agent with respect to environmental percepts.

Question Conceptual Axis: Omniscience [Knows Actual Future] vs Rationality [Maximizes Expected Utility given Past Percepts].

STEP-BY-STEP SOLUTION:

Solution Utility Horizon Vector:

Rational Action = $\text{argmax}_a E[U(s') \mid \text{Percept History } P^*]$ whereas Omniscient Action = $\text{argmax}_a U(\text{Actual } s')$.

An omniscient agent knows the actual outcome of its actions in advance (impossible in non-deterministic environments). A rational agent is realistic: it selects the action that maximizes its expected performance measure given its percept history to date.

Question 3 (Unit 1): Algorithmic Bias in Intelligent Systems

1 Mark

Identify how historical data skew leads to algorithmic discrimination in machine learning models.

Question Bias Flowchart: Skewed Historical Data → Objective Loss Minimization → Amplified Disparate Impact.

STEP-BY-STEP SOLUTION:

Solution Debiasing Intervention Model:

Input Data Pipeline [Re-sampling/Re-weighting] → Fair Loss Formulation $L_{\text{fair}} = L_{\text{ask}} + \lambda \cdot \text{Disparity}$ → Unbiased Decisions.

Supervised models minimize statistical loss over training data distributions. If historical datasets contain societal prejudices (e.g. loan approval gender disparity), the optimizer treats these disparities as genuine predictive signals, codifying systemic bias into automated policy.

Question 4 (Unit 2): Space Complexity Bottleneck of Breadth-First Search (BFS)

1 Mark

Why does Breadth-First Search exhaust physical RAM long before it exhausts allowable CPU compute time?

Question Frontier Expansion Diagram: Level 0: 1 node | Level 1: b nodes | Level d: b^d nodes kept simultaneously in memory.

STEP-BY-STEP SOLUTION:

Solution Memory Consumption Metric:

Space Complexity = $O(b^d)$. For branching factor $b=10$, depth $d=10$: $10^{10} \times 100 \text{ ext(bytes) } \approx 1 \text{ ext(Terabyte RAM)}$!

BFS must store the entire active search frontier at level d in its FIFO queue. Because memory grows exponentially as $O(b^d)$, physical RAM saturates in minutes, whereas DFS consumes only linear memory $O(b \cdot m)$.

Question 5 (Unit 2): Mathematical Condition for A* Heuristic Consistency**1 Mark**

State the triangle inequality formulation required for a heuristic to be consistent (monotonic).

Question Node Graph Triangle: Node $n \rightarrow$ [Actual Edge Cost $c(n, a, n')$] $\rightarrow$ Successor $n' \rightarrow$ [Heuristic $h(n')$] $\rightarrow$ Goal.

STEP-BY-STEP SOLUTION:

Solution Consistency Triangle Inequality:

$h(n) \leq c(n, a, n') + h(n')$ for all successors n' generated by action a .

Consistency ensures that the estimated cost $f(n)$ along any path never decreases: $f(n') = g(n') + h(n') \geq g(n) + h(n) = f(n)$. This guarantees that the first time A* visits any state, it has discovered its optimal path.

Question 6 (Unit 2): The Purpose of Skolemization in FOPC**1 Mark**

Why cannot existential quantifiers simply be dropped like universal quantifiers during CNF conversion?

Question Scope Trap: $\forall x \exists y \text{ Likes}(x, y)$ [Each x likes someone different] vs $\exists y \forall x \text{ Likes}(x, y)$ [Someone is liked by everyone].

STEP-BY-STEP SOLUTION:

Solution Skolem Function Substitution Model:

$\exists y$ bound within $\forall x$ scope $\Rightarrow$ Substitute y with Skolem Function $f(x)$: $\forall x \text{ Likes}(x, f(x))$.

Existential quantifiers assert existence of an entity that depends upon enclosing universal quantifiers. Dropping $\exists y$ naively turns it into a universal variable, destroying semantic validity. It must be replaced with a Skolem constant or function.

Question 7 (Unit 3): Explanation Facility in Expert Systems**1 Mark**

Explain how an Expert System dynamically generates an answer to a user's "HOW" query.

Question Query Flow: User prompts "HOW was Diagnosis D reached?" $\rightarrow$ Inference Engine traces active rule stack.

STEP-BY-STEP SOLUTION:

Solution Derivation Tree Traceback:

Conclusion $D \leftarrow$ Rule $R14(\text{Fact } A \wedge \text{Fact } B) \leftarrow$ Rule $R2(\text{Percept } X)$. Displays rule antecedents to user.

The Explanation Facility inspects the execution log of fired rules stored in working memory. To answer "HOW D", it displays the specific rule that concluded D along with the antecedent facts that triggered it.

Question 8 (Unit 3): Forward Chaining vs. Backward Chaining Trigger Criteria**1 Mark**

Under what condition is Backward Chaining vastly superior to Forward Chaining?

Question Search Tree Comparison: Forward Chaining [Branching Factor 100 facts $\rightarrow$ massive tree] vs Backward Chaining [1 specific goal hypothesis].

STEP-BY-STEP SOLUTION:

Solution Directed Goal Sub-tree Pruning:

Target Hypothesis $\rightarrow$ checks ONLY rules establishing Goal $\rightarrow$ prunes 95% irrelevant facts.

Backward Chaining is superior when the number of known input facts is enormous but the system needs to verify one specific hypothesis (e.g. medical diagnosis: "Does the patient have Meningitis?"), avoiding millions of irrelevant forward rule deductions.

Question 9 (Unit 3): The Knowledge Acquisition Bottleneck**1 Mark**

Define the Knowledge Acquisition Bottleneck encountered by knowledge engineers.

Question Communication Gap: Human Expert (Tacit, intuitive wisdom) $\rightarrow$ [Semantic Gap] $\rightarrow$ Knowledge Engineer (Formal IF-THEN code).

STEP-BY-STEP SOLUTION:

Solution Knowledge Engineering Protocol:

Structured Interviews + Protocol Analysis $\rightarrow$ Intermediate Conceptual Models $\rightarrow$ Formal Executable Rules.

The Knowledge Acquisition Bottleneck refers to the difficulty of extracting intuitive, unstructured domain heuristics from human experts who cannot easily articulate their internal reasoning, and translating them into formal production rules.

Question 10 (Unit 4): Limitation of Accuracy in Imbalanced Datasets**1 Mark**

Calculate the accuracy of a naive classifier that predicts all transactions as non-fraudulent on a dataset with 990 genuine and 10 fraudulent samples.

Question Class Distribution: 99% Negative Class (Genuine) | 1% Positive Class (Fraud) | Classifier predicts all Negative.

STEP-BY-STEP SOLUTION:**Solution Accuracy Paradox Formulation:**

Accuracy = $(TP + TN) / Total = (0 + 990) / 1000 = 99.0\%$. Yet Fraud Recall is $0 / 10 = 0\%$ (Complete Failure!).

Accuracy yields a deceptive score of 99.0% despite failing completely on the minority class of interest. In imbalanced domains, Precision, Recall, and F1-score must be used instead.

Question 11 (Unit 4): Why k-Nearest Neighbors is Termed a "Lazy Learner"**1 Mark**

Why is the k-NN algorithm classified as a Lazy Learner rather than an Eager Learner?

Question Training vs Inference Cost: Eager [Heavy Training, Instant Inference] vs Lazy [Zero Training, Heavy Distance Computation at Inference].

STEP-BY-STEP SOLUTION:**Solution Computation Delay Model:**

Training Time = $O(1)$ [Simply stores data] → Test Time = $O(N \cdot d)$ [Computes distances to all N points].

k-NN performs zero explicit generalization during training (it simply stores the training instances in memory). It defers all computation until query time, computing distances to every stored point upon inference.

Question 12 (Unit 4): The Apriori Principle in Frequent Itemset Mining**1 Mark**

State the anti-monotonicity property of the Apriori principle that enables search space pruning.

Question Pruning Lattice: If itemset (Beer, Diaper) is Infrequent → Superset (Beer, Diaper, Milk) is guaranteed Infrequent!

STEP-BY-STEP SOLUTION:**Solution Anti-Monotone Lattice Formulation:**

$Support(X \cup Y) \leq Support(X)$. Therefore: $Support(X) < min_sup \Rightarrow Support(X \cup Y) < min_sup$.

The Apriori Principle states that all non-empty subsets of a frequent itemset must also be frequent. Conversely, if an itemset is infrequent, all of its supersets will inevitably be infrequent, allowing entire sub-lattices to be pruned immediately.

SECTION B: 4 ANALYTICAL QUESTIONS (2 MARKS EACH = 8 MARKS)

Question 13 (Unit 1): Formal Mathematical Formulation of PEAS for an Automated Diagnostic System

2 Marks

Formulate the complete PEAS descriptor matrix for an Automated Medical Diagnostic Expert Agent.

Question Entity Relational Architecture: Patient Symptoms → Diagnostic System → Recommended Treatment Plan.

STEP-BY-STEP SOLUTION:

Solution PEAS System Grid:

Performance [Diagnostic accuracy > 99%, Recovery time minimization, Zero misdiagnosis toxicity]

Environment [Hospital ward, Pathology lab reports, Dynamic patient physiological records]

Actuators [Display monitor, Electronic prescription generator, Automated infusion pump]

Sensors [Digital thermometer, ECG sensors, Blood chemistry analyzer, Keyboard entry]

The agent optimizes its decision function $f: P^* \rightarrow A$ by mapping incoming clinical lab observations to targeted treatment regimens while strictly enforcing toxicity safety bounds.

Question 14 (Unit 2): Resolution Proof by Contradiction in Propositional Logic

2 Marks

Given Knowledge Base $KB = \{A \vee B, \neg B \vee C, \neg C\}$, prove statement A using Resolution Refutation.

Question Initial Clause Pool: Clause 1: $(A \vee B)$ | Clause 2: $(\neg B \vee C)$ | Clause 3: $(\neg C)$ | Target Goal: A .

STEP-BY-STEP SOLUTION:

Solution Resolution Refutation Tree:

Negate Goal: $C_4 = (\neg A)$

Resolve $C_2 (\neg B \vee C)$ with $C_3 (\neg C) \Rightarrow C_5 = (\neg B)$

Resolve $C_1 (A \vee B)$ with $C_5 (\neg B) \Rightarrow C_6 = (A)$

Resolve $C_6 (A)$ with $C_4 (\neg A) \Rightarrow \square$ (EMPTY CLAUSE / CONTRADICTION!)

Derivation of the empty clause $\square$ proves that negating A leads to a logical contradiction with the KB. Therefore, statement A is proven true by refutation.

Question 15 (Unit 3): Certainty Factor (CF) Propagation in Rule-Based Systems

2 Marks

In MYCIN, if evidence E has certainty $CF(E) = 0.8$ and Rule R is "IF E THEN H ($CF = 0.7$)", compute the final hypothesis certainty $CF(H, E)$.

Question Evidence-to-Hypothesis Flow: Evidence E [$CF = 0.8$] → Rule R [Attenuation Factor $CF_{rule} = 0.7$] → Hypothesis H .

STEP-BY-STEP SOLUTION:

Solution MYCIN Certainty Multiplication Model:

$CF(H, E) = CF(E) \times CF_{rule} = 0.8 \times 0.7 = 0.56$.

When an antecedent fact is not known with absolute certainty, MYCIN scales the rule's confidence by the evidence certainty. Here, the final certainty in hypothesis H is exactly 0.56 (56% confidence).

Question 16 (Unit 4): Complete Confusion Matrix Computation

2 Marks

A cancer diagnostic test yields $TP = 80, FP = 20, FN = 10, TN = 890$. Calculate Precision, Recall, and F1-Score.

Question 2x2 Contingency Matrix: Actual Positives = 90 ($80 TP + 10 FN$) | Actual Negatives = 910 ($20 FP + 890 TN$).

STEP-BY-STEP SOLUTION:

Solution Performance Metrics Derivation:

Precision = $TP / (TP + FP) = 80 / (80 + 20) = 80 / 100 = 0.80$ (80.0%)

Recall = $TP / (TP + FN) = 80 / (80 + 10) = 80 / 90 = 0.8889$ (88.89%)

F1-Score = $2 \times \frac{0.80 \times 0.8889}{0.80 + 0.8889} = 1.4222 / 1.6889 = 0.8421$ (84.21%)

The model captures ~88.9% of all true cancer cases while maintaining an 80.0% precision rate, resulting in an balanced F1-score of 0.8421.

SECTION C: 4 COMPREHENSIVE PROBLEMS (4 MARKS EACH = 16 MARKS)

Question 17 (Unit 1): Complete Architecture and Internal Control Loop of Learning Agents

4 Marks

Illustrate and explain the detailed four-component architectural decomposition of a Learning Agent.

Question Architectural Feedback Loop: Environment → Sensors → Critic & Performance Element → Learning Element → Problem Generator → Actuators.

STEP-BY-STEP SOLUTION:

Solution 4-Component Decomposition Model:

- Critic: Evaluates agent performance against fixed external standard; generates scalar feedback
- Learning Element: Responsible for making improvements; adjusts internal policy weights
- Learning Goals: Specifies target quality standards to the critic
- Problem Generator: Suggests exploratory actions leading to novel perceptual experiences

The **Performance Element** is the traditional agent selecting actions based on percepts. The **Critic** compares real outcomes against performance standards and feeds errors to the **Learning Element**. Crucially, the **Problem Generator** prevents the agent from getting trapped in local sub-optimal routines by deliberately proposing exploratory actions to discover better policies.

Question 18 (Unit 2): Step-by-Step Execution of A* Search on a Weighted Graph

4 Marks

Trace A* search from Start node S to Goal node G. Edge costs: S-A=2, S-B=4, A-C=3, A-D=5, B-D=2, C-G=4, D-G=3. Heuristics: h(S)=7, h(A)=5, h(B)=4, h(C)=3, h(D)=2, h(G)=0.

Question Graph Topology: S → A (cost 2) & B (cost 4); A → C (cost 3) & D (cost 5); B → D (cost 2); C → G (cost 4); D → G (cost 3).

STEP-BY-STEP SOLUTION:

Solution Priority Queue Frontier Execution Trace:

Step 1: Init Queue = [S(f=0+7=7)] → Pop S

Expand S → A: g=2, f=2+5=7; B: g=4, f=4+4=8. Queue = [A(f=7), B(f=8)] → Pop A

Expand A → C: g=2+3=5, f=5+3=8; D: g=2+5=7, f=7+2=9. Queue = [B(f=8), C(f=8), D(f=9)] → Pop B

Expand B → D: g=4+2=6, f=6+2=8 (Cheaper D path found!). Queue = [C(f=8), D(f=8)] → Pop C

Expand C → G: g=5+4=9, f=9+0=9. Queue = [D(f=8), G(f=9)] → Pop D

Expand D → G: g=6+3=9, f=9+0=9. Queue = [G(f=9)] → Pop G (GOAL REACHED!)

Optimal Path Found: S → B → D → G with total optimal path cost = 4 + 2 + 3 = 9. (Path S-A-C-G also has cost 9; A* correctly guarantees discovery of optimal cost).

Question 19 (Unit 3): Conflict Resolution Strategies in Expert System Production Rules

4 Marks

Explain the Conflict Resolution problem when multiple rules match working memory simultaneously. Discuss Refractoriness, Recency, and Specificity.

Question Conflict Set: Working Memory (Fact 1, Fact 2) → Matches Rule 1, Rule 5, Rule 12 → Which rule fires?

STEP-BY-STEP SOLUTION:

Solution Conflict Resolution Engine Model:

- Specificity: Fire rule with most antecedents (IF A ∧ B ∧ C fires over IF A)
- Recency: Fire rule matching most recently asserted facts in Working Memory
- Refractoriness: Never fire same rule instance twice on identical data (prevents loops)
- Rule Priorities / Weights: Pre-assigned static salience integers

Without conflict resolution, the inference engine enters infinite loops or applies broad generic rules instead of domain-specific exceptions. Applying Specificity ensures specialized knowledge takes precedence over general defaults.

Question 20 (Unit 4): Information Gain Computation for Decision Tree Splitting**4 Marks**

A dataset has 14 samples: 9 Positive (+), 5 Negative (-). An attribute "Wind" splits into: Weak (6+, 2-) and Strong (3+, 3-). Compute Information Gain: Gain(S, Wind).

Question Partition Tree: Root S [9+, 5-] → Weak [6+, 2-] (8 samples) and Strong [3+, 3-] (6 samples).

STEP-BY-STEP SOLUTION:**Solution Entropy & Gain Derivation:**

$$\text{Entropy}(S) = - \frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.9403 \text{ bits}$$

$$\text{Entropy}(S_{\text{Weak}}) = - \frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.8113 \text{ bits}$$

$$\text{Entropy}(S_{\text{Strong}}) = - \frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0000 \text{ bits}$$

$$\text{Expected Entropy} = \frac{8}{14} \times 0.8113 + \frac{6}{14} \times 1.0000 = 0.4636 + 0.4286 = 0.8922 \text{ bits}$$

$$\text{Gain}(S, \text{Wind}) = 0.9403 - 0.8922 = 0.0481 \text{ bits}$$

Attribute "Wind" reduces disorder by **0.0481 bits**. The decision tree compares this Gain against all remaining candidate attributes and selects the maximum for the root split.

SECTION D: 4 EXTENDED ESSAY & DESIGN PROBLEMS (6 MARKS EACH = 24 MARKS)

Question 21 (Unit 1): Complete Architectural Comparison of 5 Core Intelligent Agent Classes 6 Marks

Provide the structural internal architectures, operational rules, advantages, and terminal failure conditions for Simple Reflex, Model-Based, Goal-Based, Utility-Based, and Learning Agents.

Question Agent Evolution Continuum: Condition-Action Lookup → Internal State Tracking → Multi-Step Goal Planning → Utility Trade-offs → Adaptive Self-Improvement.

STEP-BY-STEP SOLUTION:

Solution 5-Agent Architecture Comparison Matrix:

- 1. Simple Reflex: Sensor → Condition-Action Rule → Actuator. Fails completely in partially observable worlds
- 2. Model-Based: Maintains internal state ("How the world evolves" + "How my actions affect world")
- 3. Goal-Based: Combines State + Goals; uses search/planning to achieve target states
- 4. Utility-Based: Evaluates trade-offs using continuous Utility Function U(s) when goals conflict
- 5. Learning Agent: Decoupled into Performance Element, Critic, Learning Element, and Problem Generator

Detailed Analysis: Simple reflex agents are paralyzed by unobserved environmental features because their mapping $f: P \rightarrow A$ ignores history. Model-based agents overcome this by preserving a hidden state vector. Goal-based agents add planning foresight but treat success as a binary variable. Utility-based agents map states to real numbers, allowing rational trade-offs (e.g. trading speed for passenger safety). Learning agents adapt their decision boundaries continuously from experience.

Question 22 (Unit 2): Full Propositional and Predicate Logic Formal Proof Pipeline 6 Marks

Convert the following facts to FOPC, convert to CNF, and prove using Resolution: (1) "Marcus was a man", (2) "Marcus was a Pompeian", (3) "All Pompeians were Romans", (4) "Caesar was a ruler", (5) "All Romans were either loyal to Caesar or hated him", (6) "Everyone is loyal to someone", (7) "Men only cause trouble to people they do not like". Prove: "Marcus hated Caesar".

Question Knowledge Base Clauses: $Man(Marcus)$, $Pompeian(Marcus)$, $\forall x \text{ Pompeian}(x) \Rightarrow Roman(x)$, $\forall x \text{ Roman}(x) \Rightarrow (Loyal(x, Caesar) \vee Hate(x, Caesar))$.

STEP-BY-STEP SOLUTION:

Solution Full CNF & Resolution Pipeline:

C1: $Man(Marcus)$
C2: $Pompeian(Marcus)$
C3: $\neg Pompeian(x) \vee Roman(x)$
C4: $\neg Roman(y) \vee Loyal(y, Caesar) \vee Hate(y, Caesar)$
C5: $\neg Loyal(Marcus, Caesar)$ [Asserted from context facts]
Negate Target: C6: $\neg Hate(Marcus, Caesar)$
Resolve C2 with C3 (subst (x/Marcus)) $\Rightarrow$ C7: $Roman(Marcus)$
Resolve C7 with C4 (subst (y/Marcus)) $\Rightarrow$ C8: $Loyal(Marcus, Caesar) \vee Hate(Marcus, Caesar)$
Resolve C8 with C5 $\Rightarrow$ C9: $Hate(Marcus, Caesar)$
Resolve C9 with C6 $\Rightarrow \square$ (EMPTY CLAUSE DERIVED!)

The successful unification and resolution to the empty clause $\square$ provides rigorous, sound proof that **Marcus hated Caesar**.

Question 23 (Unit 3): Complete Engineering Design of a Medical Expert System 6 Marks

Design an Expert System for Infectious Disease Diagnosis. Specify the Knowledge Representation format, Inference Chaining Strategy, Explanation Dialogue, and Certainty Combination rules.

Question Clinical Diagnostic Architecture: Patient Bloodwork → Rule Pattern Matcher → Working Memory → Explanation Generator.

STEP-BY-STEP SOLUTION:

Solution Multi-Tier Production System Design:

- 1. KB Rule Formalism: IF (Culture = GramNegative) $\wedge$ (Morphology = Rod) $\wedge$ (Aerobicity = Anaerobic) THEN (Organism = Bacteroides [CF = 0.85])
- 2. Inference: Backward Chaining from suspected critical pathogens; triggers Forward Chaining on emergency vital flags
- 3. Combining Uncertainties: $CF_{comb}(CF_1, CF_2) = CF_1 + CF_2 - (CF_1 \times CF_2)$ for positive certitudes
- 4. Explanation Engine: Implements execution-stack unwinding for user inquiries

This system decouples patient data from clinical disease rules. By maintaining certainty factors, it safely handles partial and ambiguous lab cultures. The explanation engine generates trust among clinicians by detailing the exact evidence trail behind every prescribed antibiotic regimen.

Given 1D data points $X = (2, 4, 10, 12, 3, 20, 30, 11, 25)$. Cluster into $k=2$ groups using initial centroids $\mu_1 = 2$ and $\mu_2 = 4$. Perform complete iterations until convergence.

Question Centroid Displacement Vector: $\mu_1 = 2, \mu_2 = 4 \rightarrow$ Assign all points $\rightarrow$ Recalculate means $\rightarrow$ Check $\Delta\mu = 0$.

STEP-BY-STEP SOLUTION:

Solution Iterative Centroid Convergence Trace:

Iteration 1 Assignment:

Distance to $\mu_1(2)$ vs $\mu_2(4)$:

Point 2: $|2-2|=0 \leq |2-4|=2 \Rightarrow$ Cluster 1

Point 3: $|3-2|=1 = |3-4|=1 \Rightarrow$ Cluster 1 (tie break)

Points (4, 10, 11, 12, 20, 25, 30) all closer to $\mu_2=4 \Rightarrow$ Cluster 2

Cluster 1 = (2, 3) $\Rightarrow$ New Centroid $\mu_1' = (2+3)/2 = 2.5$

Cluster 2 = (4, 10, 11, 12, 20, 25, 30) $\Rightarrow$ New Centroid $\mu_2' = 112 / 7 = 16.0$

Iteration 2 Assignment ($\mu_1=2.5, \mu_2=16.0$):

Points (2, 3, 4) closer to 2.5 $\Rightarrow$ Cluster 1

Points (10, 11, 12, 20, 25, 30) closer to 16.0 $\Rightarrow$ Cluster 2

New Centroid $\mu_1'' = (2+3+4)/3 = 3.0$

New Centroid $\mu_2'' = (10+11+12+20+25+30)/6 = 108 / 6 = 18.0$

Iteration 3 Assignment ($\mu_1=3.0, \mu_2=18.0$):

Points (2, 3, 4, 10) closer to 3.0? $|10-3|=7$ vs $|10-18|=8 \Rightarrow$ Point 10 moves to Cluster 1!

Cluster 1 = (2, 3, 4, 10) $\Rightarrow \mu_1''' = 19 / 4 = 4.75$

Cluster 2 = (11, 12, 20, 25, 30) $\Rightarrow \mu_2''' = 98 / 5 = 19.6$

Iteration 4 Assignment ($\mu_1=4.75, \mu_2=19.6$):

Cluster 1 = (2, 3, 4, 10, 11, 12); Cluster 2 = (20, 25, 30)

$\mu_1 = 42/6 = 7.0$; $\mu_2 = 75/3 = 25.0$

Iteration 5 Assignment: Assignments remain identical! NO SHIFT $\Rightarrow$ CONVERGENCE ACHIEVED!

Final Converged Clusters:

• **Cluster 1:** (2, 3, 4, 10, 11, 12) with final mean centroid $\mu_1 = 7.0$.

• **Cluster 2:** (20, 25, 30) with final mean centroid $\mu_2 = 25.0$.